

Particles and Nuclei 2

- Third Day 1/29/26
- Announcements:
- Canvas
- HW 2: Released Tonight Due Thursday 2 wks from today in Class
- READING Lancaster & Blundell (L&B) Ch ~6 & 7, 4, [handout pdf, slightly less than that]: Finish Ch 4 eventually, but wait

Rel. Kine. Starting Point E, p

- Relations btw Energy and Momentum p change for Rel.

$$E = mc^2 \text{ (at rest)}$$

$$E^2 = \vec{p}^2 + m^2$$

$$E = \pm \sqrt{p^2 + m^2}$$

$$E^2 - \vec{p}^2 = m^2$$

$$\begin{aligned} E^2 &= \vec{p}^2 \\ E &= |\vec{p}| \\ p &\gg m \\ &\text{(not } p^2/2m) \end{aligned}$$

New Lorentz Transf. Equations

- Lorentz Transformations (in particle physics) key equations are now for p & E not in terms of x, t —and leaving out the c 's :
- Use $p_{\parallel}$ instead of p_x or p_z : in particle physics, usually z is preferred as “typical” boosting direction but e.g. Griffiths uses x , like most elementary treatments

$$p_{\parallel}' = \gamma (p_{\parallel} \pm \beta E)$$
$$E' = \gamma (E \pm \beta p_{\parallel})$$

$\pm$ depends what you're trying to do
formulas are almost always quoted as in Griff
for both transformation or “reverse transformation”
which is which?

THIS WILL BE ONE FOCUS OF TODAY

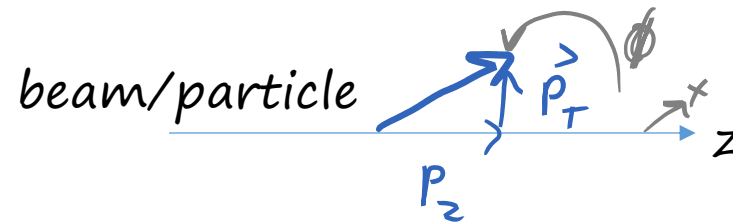
Relativistic Kinematics - Basics

- Energy-mom $(E, \vec{p}) = p^\mu$
- Time Space ~~$(t, \vec{x}) = x^\mu$~~
- Velocity ~~$(\gamma, \gamma\vec{B}) = v^\mu$~~

$$\vec{p} = (p_x, p_y, p_z) \\ = (\vec{p}_T, p_z)$$

$$\vec{\beta} = \frac{\vec{v}}{c}$$

(not used)



Scalar Product

$$p_1 \cdot p_2 = p_1^\mu \cdot p_{2\mu} = p_1^\mu p_2^\nu g_{\mu\nu}$$

$$p^2 = p \cdot p = E^2 - |\vec{p}|^2 \\ = m^2 !!!$$

$$\mathbf{A} \cdot \mathbf{B} = (A^0 \ A^1 \ A^2 \ A^3) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} B^0 \\ B^1 \\ B^2 \\ B^3 \end{pmatrix}$$

Lorentz Transformations

$$\gamma = (1 - \beta^2)^{-1/2}$$

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} \quad \Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$p'^{\mu} = \Lambda^{\mu}_{\nu} p^{\nu}$$

$$p_{\mu} = (E, -\vec{p})$$

$$p'^{\mu} = \Lambda^{\mu}_{\nu} p^{\nu}$$

Einstein summation?

Some Numerical Things

- A few values of β, γ

- p or $E \approx 2m \rightarrow \gamma = 2 \quad \beta^2 = 3/4 \rightarrow \beta = 0.86$

- p or $E \approx 3m \rightarrow \gamma = 3 \quad \beta^2 = 8/9 \rightarrow \beta = 0.95$

- p or $E \approx 4m \rightarrow \gamma = 4 \quad \beta^2 = 15/16 \rightarrow \beta = 0.97$

(halfway btw
0.75 & 1 *)

$$\gamma^2 = \frac{1}{1-\beta^2}$$

$$\beta^2 = \frac{\gamma^2 - 1}{\gamma^2} = 1 - 1/\gamma^2$$

This is an often used simplifier algebraically but it's good to remember in this numerical way

* $\beta = \text{sqrt}(\beta^2) = \text{sqrt}(1-\epsilon) \approx 1 - (1/2)\epsilon \quad [\epsilon = 1/\gamma^2]$

Some Numerical Things

- A few values of β, γ

- p or $E \approx 2m \rightarrow \gamma = 2 \quad \beta^2 = 3/4 \rightarrow \beta = 0.86$

- p or $E \approx 3m \rightarrow \gamma = 3 \quad \beta^2 = 8/9 \rightarrow \beta = 0.95$

- p or $E \approx 10m \rightarrow \gamma = 10 \quad \beta^2 = 99/100 \rightarrow \beta = 0.995$

(halfway btw
0.75 & 1 *)

$$\gamma^2 = \frac{1}{1-\beta^2}$$

$$\beta^2 = \frac{\gamma^2 - 1}{\gamma^2} = 1 - 1/\gamma^2$$

This is an often used simplifier algebraically but it's good to remember in this numerical way

$$* \quad \beta = \text{sqrt}(\beta^2) = \text{sqrt}(1-\epsilon) \approx 1 - (1/2)\epsilon \quad [\epsilon = 1/\gamma^2]$$

In reverse

- Perhaps more useful is using this numerical trick to go backwards.
- If $\beta = 0.99995$ what's gamma? (quickly without plugging into a computer?)
 - Order of magnitude : $1 \sim e^{-4}$ less than 1... β^2 is too, $1 \sim e^{-4} = 1/\gamma^2$
... $1e+4 = \gamma^2$ $\gamma = 1e2$ so $\text{sqrt}(1/\text{diff}(\beta^2 \text{ from } 1))$
 - $2 * 0.00005 \rightarrow = 0.0001 \rightarrow \beta^2 = 0.9999 = 9999/10000$
 - $-\gamma^2 = 10000$, $\gamma = 100$

Relativistic Kinematics – Basics

- γ factor $\rightarrow$ size of Spec. Rel. effects $\gamma \geq 1$
- Time Dilation (times γ)
- Length Contraction (divided by γ)
- USUALLY ANOTHER PARTICLE IS BOOST FRAME (“Beta”)

$$\beta = \frac{p}{E}$$

$$\gamma = \frac{E}{m}$$

- Good application cosmic ray muons – lifetime $2.2 \mu\text{s}$
- $c\tau = 660 \text{ m}$
- spectra of cosmic rays ($\sim$ not muons in space) from space average/peak at 4 GeV (high energy—even on the surface of the Earth!)
- Gamma factor (?)
- ≈ 40 can travel 20 km
- (ignoring μ energy loss – 2 GeV per $\sim 15 \text{ km}$ air)
- Lifetime dilation like this: the most visible proof of SR

Relativistic Kinematics – Basics

- γ factor $\rightarrow$ size of Spec. Rel. effects $\gamma \geq 1$
- Time Dilation (times γ)
- Length Contraction (divided by γ)
- USUALLY ANOTHER PARTICLE IS BOOST FRAME (“Beta”)

$$\beta = \frac{p}{E}$$

$$\gamma = \frac{E}{m}$$

$$\gamma \beta = \frac{p}{m}$$

Interesting/sometimes helpful to write these boost variables in terms of these “boost frame particle” E_b, p_b

Boost Matrix 4v

$$\Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- interesting to look at in terms of Boost particle frame compont's of p^μ

$$= \frac{1}{m} \begin{pmatrix} E_b & -p_b & 0 & 0 \\ -p_b & E_b & & \\ & & m & \\ & & & m \end{pmatrix}$$

- dfd Also can see more explicitly why inverse matrix is same w/o the minus signs $\rightarrow$ by symm $\vec{p}_b$ must flip signs

Transf. or Inverse Transf?

- Same question: but is the minus sign for going TO the transformed frame or FROM the transformed frame?
 - Can be hard to remember – even deduce:

$$\Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Inverse transformation (WITHOUT minus sign) is one people actually mostly use and remember.
- We call it a BOOST!!!! (... Terminology:)

- “BOOST” (+ signs) like an “active rotation”
- Transformation: like (passive) rotation transf operator e^{-iR}
 - (remember the minus sign on the exponential?)

Transf. or Inverse Transf?

- Same question: but is the minus sign for going TO the transformed frame or FROM the transformed frame?
 - Can be hard to remember – even deduce:

$$\Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Inverse transformation (WITHOUT minus sign) is one people actually mostly use and remember.
- We call it a BOOST!!!! (... Terminology:)

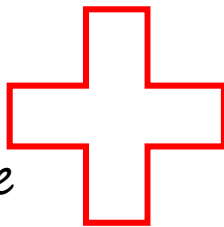
- “BOOST” (+ signs) like an “active rotation”
- Transformation: like (passive) rotation transf operator e^{-iR}
 - (remember the minus sign on the exponential?)

Boost vs Transformation

- Most Lorentz / Four Vector code packages implement a “four-vector”, with a “Boost” boost function
- E.g. `root.cern.ch` `TLorentzVector` class (obj)
 - `TLorentzVector::Boost(4vec);` `TLV myLv; myLV.Boost(...);`
- `CLHEP/ HEPMC` : `#include "CLHEP/Vector/LorentzVector.h"`

Boost: Particle Decay:
(active)

Decay in COM frame



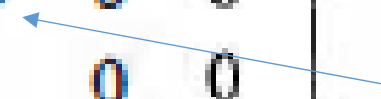
Kinematics then “Boosted” to
particles momentum frame

Transform: Momentum of
particle a in frame of moving
particle b. (passive)

 sign

Transf. or Inverse Transf?

- Same question: but is the minus sign for going TO the transformed frame or FROM the transformed frame?
 - Can be hard to remember – even deduce:

$$\Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$


- Inverse transformation (WITHOUT minus sign) is one people actually mostly use and remember.
- We call it a BOOST!!!! (... Terminology:)

- “BOOST” (+ signs) like an “active rotation”
- Transformation: like (passive) rotation transf operator e^{-iR}
 - (remember the minus sign on the exponential?)

Transf. or Inverse Transf?

- Same question: but is the minus sign for going TO the transformed frame or FROM the transformed frame?

- Can be hard to remember – even deduce:

$$\Lambda = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In the
normal
transform
logic $\beta \equiv \vec{\beta}$
here
MUST be
a signed
1D vector

- Inverse transformation (WITHOUT minus sign) is one people actually mostly use and remember.
- We call it a BOOST!!!! (... Terminology:)

- “BOOST” (+ signs) like an “active rotation”
- Transformation: like (passive) rotation transf operator e^{-iR}
 - (remember the minus sign on the exponential?)

BOOST or Transformation?

- Same question: but is the minus sign for going TO the transformed frame or FROM the transformed frame?

- Can be hard to remember – even deduce:

So Boost is

$$\Lambda_{\text{Boost}} = \begin{bmatrix} \gamma & +\gamma\beta & 0 & 0 \\ +\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Inverse transformation (WITHOUT minus sign) is one people actually mostly use and remember.
- We call it a BOOST!!!! (... Terminology:)

- “BOOST” (+ signs) like an “active rotation”
- Transformation: like (passive) rotation transf operator e^{-iR}
 - (remember the minus sign on the exponential?)

Boosts : Think of Decay of Moving Particle

- Start with Boost

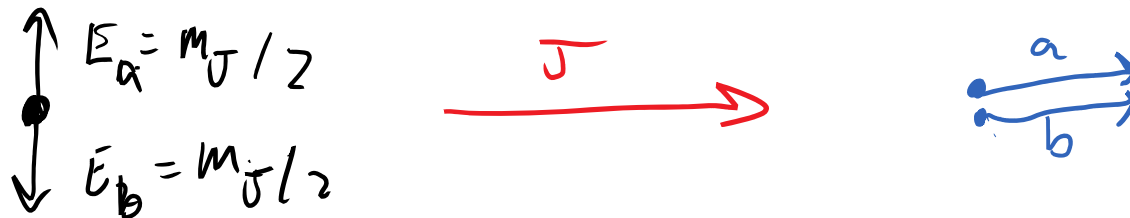
(and consider only p equation)

$$p_{||}' = \gamma (p_{||} + \beta E)$$

$$E' = \gamma (E + \beta p_{||})$$

only the + sign now: boost ie boosts

- Suppose a particle J is moving ~highly relativistically
- Suppose the particle decays into two same mass particles 1,2
- E.g. $\pi^0 \rightarrow 2\gamma$ $J/\Psi \rightarrow e+e^-$ $\Phi \rightarrow K+K^-$
- COM frame “back to back” is BOOSTED into/by J's frame



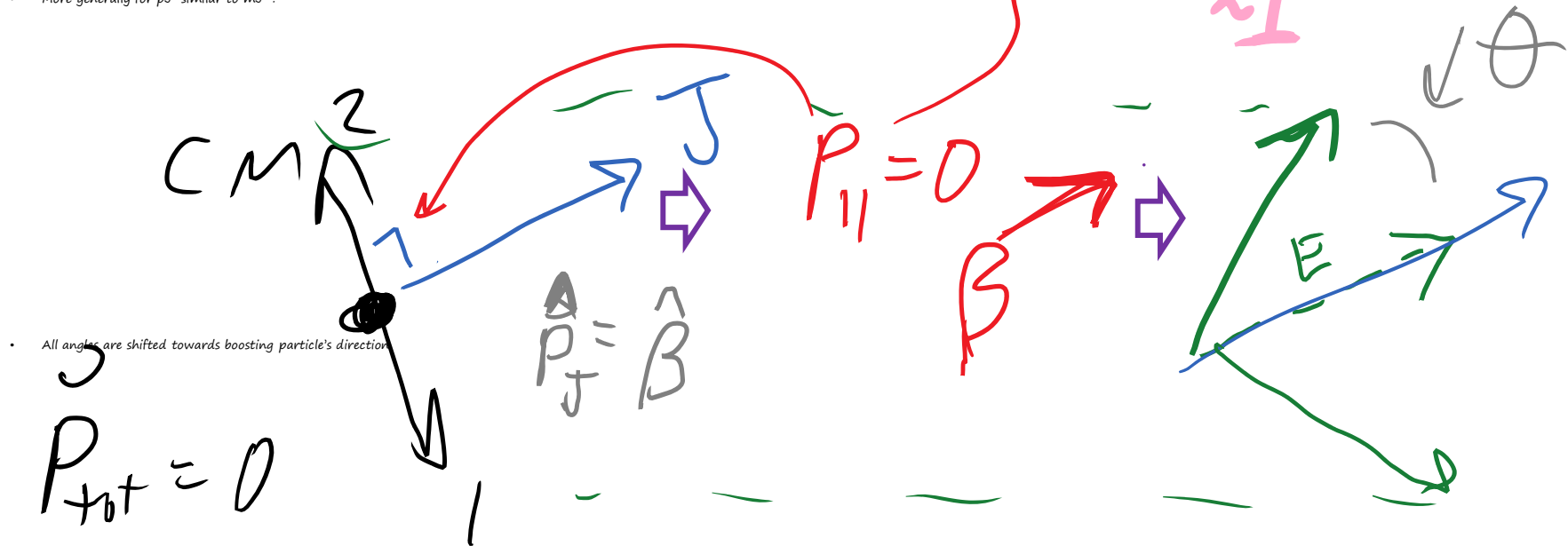
- Learn: All angles are shifted towards boosting particle's direction
- Make sense if $p_J \rightarrow \infty$ single p_J in lab must stay in same direction (decay conserves system momentum) – small mass
 - p_{tot} can't change direction much even if now two particles: $m_J / 2 \ll p_J$

Boosts : Think of Decay of Moving Particle

- *Start with Boost*

$$P_{11}' = \gamma (P_{11} + \beta E)$$

- More generally for pJ similar to mJ :



$$p_{||} = 0 \quad \Rightarrow \quad p'_{||} = \gamma m_{\sigma}/2$$

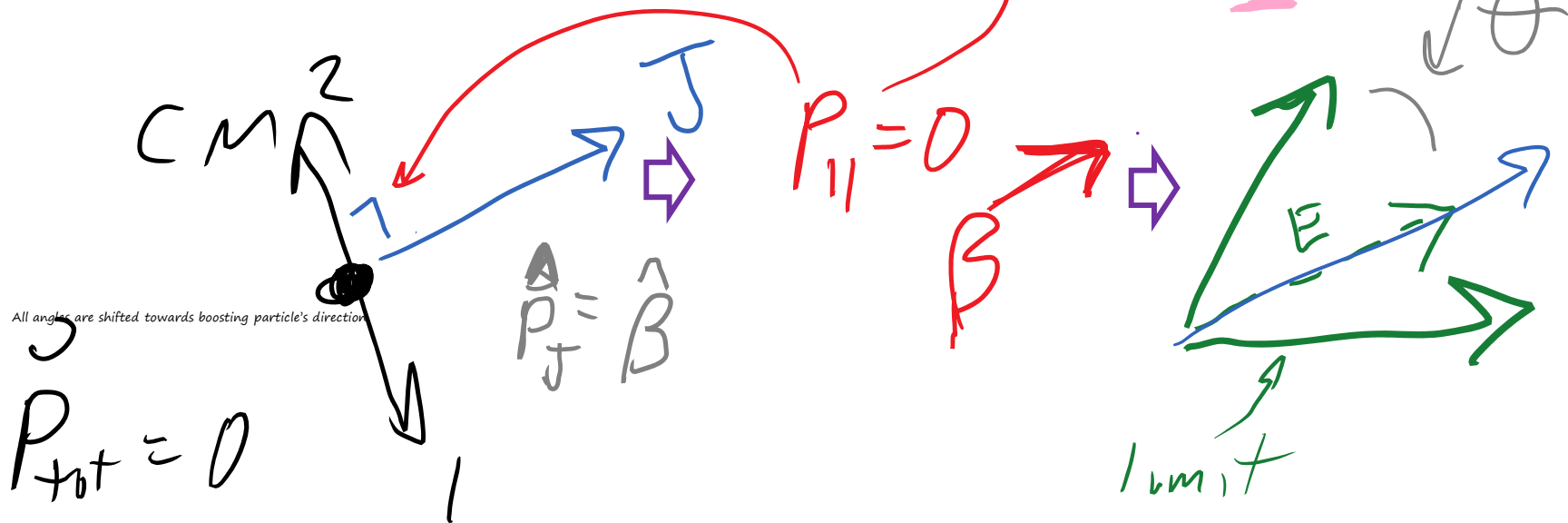
Boosts : Think of Decay of Moving Particle

- Start with Boost

$$P_{11}' = \gamma (\cancel{P_{11}} + \beta E)$$

- More generally for p_J similar to m_J :

- All angles are shifted towards boosting particle's direction



$$P_{11} = 0 \Rightarrow P_{11}' = \gamma m_{\sigma}/2$$

$$P_T' = P_T$$

The point: Angles "close" in direction of boost, but limited by initial pT's

Transformations – Invariants I

- Can find many code packages out on the internet for doing this.
- ROOT: TLorentzVector.h
- HEPMC :
- “Boost”
- **EXERCISE:**
- Find class and methods to perform Lorentz transformations.
- IN tophat give an answer in the form “classname.function()” for doing the transformation in code and also what parameters /arguments are needed for the function()
- <https://pypi.org/project/pylorentz/>
- Root -- Transformation objects/classes, as well as methods on TLorentzvector

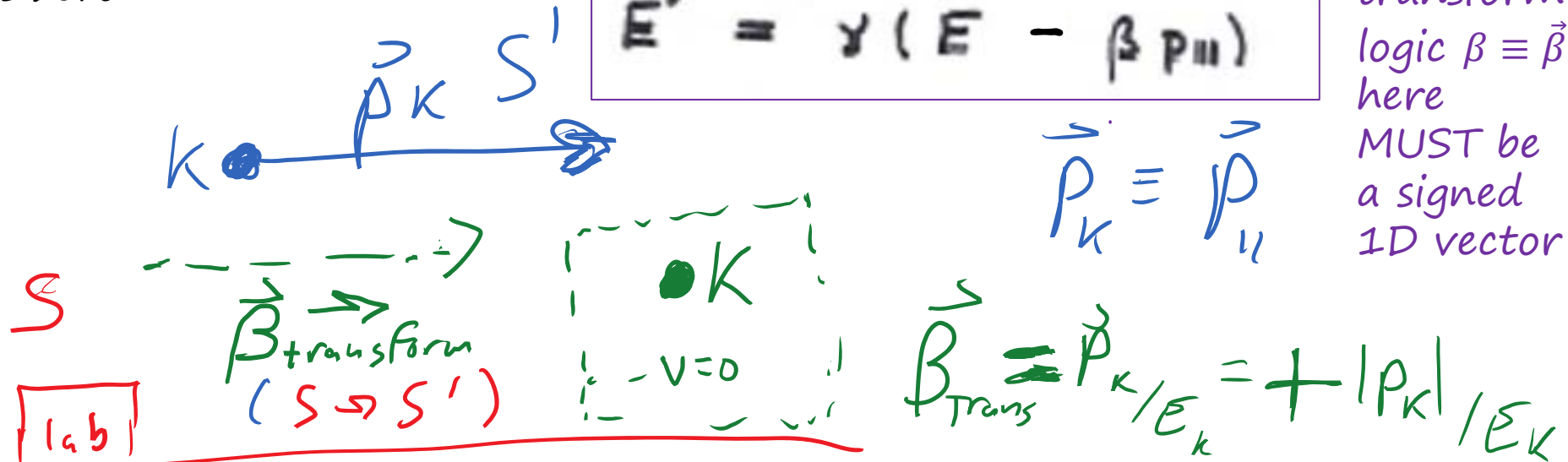
Transform : Particle to its OWN COM

- Single particle COM is frame where $\vec{p}_{\text{particle}} = 0$ is zero

$$p_{||}' = \gamma (p_{||} - \beta E)$$

$$E' = \gamma (E - \beta p_{||})$$

In the normal transform logic $\beta \equiv \vec{\beta}$ here MUST be a signed 1D vector



- The starting reference frame is the lab frame in which $\vec{p}_K$ is defined. In that frame K is moving with momentum $\vec{p}_K$ (unprimed)
- We want to transform to K 's COM frame.
 - (in which the momentum should obviously be 0)
- The relative $\vec{\beta}$ of the frame to transform to, compared to the reference (ground/lab) frame, is POSITIVE $+|\beta_K|$ (rightward)
- Obviously we need the minus sign transformations to cause the cancelation to 0 in the $p_{||}$ equation...
- ...And in the normal logic the minus sign is consistent with the convention that the minus sign version of the matrix is for normal transformations

Transform : OWN COM : Use BOOST

- Single particle COM is frame where $p_{||}$ is zero



$$p_{||}' = \gamma (p_{||} - \beta E)$$

$$E' = \gamma (E - \beta p_{||})$$

$$\vec{p}_k \equiv \vec{p}_{||}$$

- Go to COM frame i.e. not moving
- Obviously need to go "backwards" undo momentum – why?
 - ~~relative motion, negative sign? (?)~~
 - NO just need NEGATIVE BOOST!!!!
 - So better to think of transforms as Negative boost!

$$\vec{\beta} = - \frac{\vec{p}_k}{E_k} \quad \left(-|\vec{p}_k|/E \right)$$

$$p_{||}' = \gamma (p_{||} + \beta E)$$

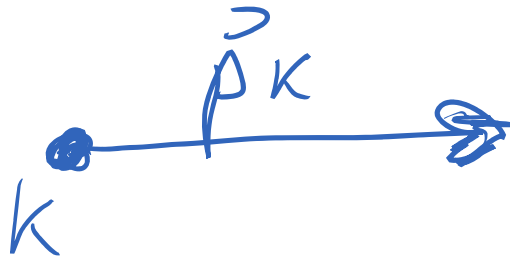
$$E' = \gamma (E + \beta p_{||})$$

Transform : Transform Moving Particle to it's OWN COM

- In other words, change K's frame to yours...

$$p_{11}' = \gamma (p_{11} + \vec{\beta} E)$$

$$E' = \gamma (E + \vec{\beta} p_{11})$$



$$\vec{p}_K \equiv \vec{p}_{11}$$

- Boost Negatively ! to change the frame from the to be transformed to "your frame".
- So COM transform is:

$$\vec{\beta} = -\frac{\vec{p}_K}{E_K}$$

$$p_{11}' = p_{11}'' = \gamma (p_{11} + \vec{\beta} E) = \gamma \left(p_{11} - p_{11} \left(\frac{E}{E} \right) \right) = 0$$

Lorentz ~~Transformations~~ - Nonmatrix

• $\vec{p}_{\perp}'' = \vec{p}_{\perp}$ (BOOST) $p_{||} = \frac{\vec{\beta} \cdot \vec{p}}{|\vec{\beta}|}$

$$\vec{p}'' = \vec{p} + \gamma \vec{\beta} \left(\frac{\gamma}{\gamma+1} \vec{\beta} \cdot \vec{p} + E \right) \quad E'' = \gamma (E + \vec{\beta} \cdot \vec{p})$$

$\underbrace{\frac{\gamma}{\gamma+1} \vec{\beta} \cdot \vec{p}}_{\sim 1?} \underbrace{\vec{\beta}}_{p_{||} \beta}$

$\downarrow$
 $\sim |\vec{p}| + p_{||}$

These formula are "boosts": i.e. for transformations w/ velocity $-\beta$ wrt to original frame

Thus they have the reverse of the usual minus sign transf convention

Double prime is LAB (boosted) frame !!! p and E refer to original frame (e.g. CoM for boosted decays)

$\perp$

~~Lorentz Transformations~~ - Nonmatrix

(BOOST)

$$\vec{p}'' = \vec{p} + \gamma \vec{\beta} \left(\frac{\gamma}{\gamma+1} \vec{\beta} \cdot \vec{p} + E \right) \quad E = \gamma(E + \vec{\beta} \cdot \vec{p})$$

Don't have to use the P_{parallel}/p_T coordinates – any axis choice

Good for coding

Similar Transformation Equation

- Wikipedia has a similar formula, but w/ using a unit vector $\mathbf{n}$ instead of $\vec{\beta}$ [$\mathbf{n} \equiv \hat{\mathbf{n}} \equiv \hat{\beta} \equiv \hat{p}_{\parallel}$]

Transformation of other quantities [\[edit\]](#)

In general, given four quantities A and $\mathbf{Z} = (Z_x, Z_y, Z_z)$ and their Lorentz-boosted counterparts A' and $\mathbf{Z}' = (Z'_x, Z'_y, Z'_z)$, a relation of the form

$$A^2 - \mathbf{Z} \cdot \mathbf{Z} = A'^2 - \mathbf{Z}' \cdot \mathbf{Z}'$$

implies the quantities transform under Lorentz transformations similar to the transformation of spacetime coordinates;

$$A' = \gamma \left(A - \frac{v \mathbf{n} \cdot \mathbf{Z}}{c} \right),$$
$$\mathbf{Z}' = \mathbf{Z} + (\gamma - 1)(\mathbf{Z} \cdot \mathbf{n})\mathbf{n} - \frac{\gamma A v \mathbf{n}}{c}.$$

From Bill Zajc Notes – Long time PHENIX Spokesperson

PRACTICAL LORENTZ KINEMATICS

I. YOU ALL KNOW THAT

$$p_{||}' = \gamma (p_{||} + \beta E)$$

$$E' = \gamma (E + \beta p_{||}) \quad , \quad \gamma \equiv \frac{1}{\sqrt{1-\beta^2}}$$

$$E = \sqrt{p^2 + m^2}$$

NOW PLEASE FORGET THIS

Instead...

- Instead...

II. PLEASE REMEMBER

A. $p \equiv (E, \vec{p})$ is a 4-vector (& masses are usually ~ 0 !)

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$D. \begin{cases} p_{||} = m_T \sinh y \\ E = m_T \cosh y \end{cases} \Leftrightarrow \begin{cases} m_T = \sqrt{m^2 + p_T^2} \\ y = \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{cases}$$

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

Rapidity

Instead...

- Instead...

II. PLEASE REMEMBER

A. $p \equiv (E, \vec{p})$ is a 4-vector (& masses are usually ~ 0 !)

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$\left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

Instead...

- Instead...

II. PLEASE REMEMBER

A. $p \equiv (E, \vec{p})$ is a 4-vector (& masses are usually ~ 0 !)

B. Products of 4-vectors are invariant

Griffiths

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$\left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

$$E. \quad \beta = p/E = \tanh y$$

$$F. \quad E \frac{d^3 n}{dp^3} = \frac{d^3 n}{dy d^2 p_T} = \text{invariant}$$

G. Lorentz transforms additive in rapidity

Instead...

- Instead...

II. PLEASE REMEMBER

A. $p \equiv (E, \vec{p})$ is a 4-vector (& masses are usually ~ 0 !)

B. Products of 4-vectors are invariant

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$\left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

Needs the
boost/transf
equation

Very important
We need to
define invariant
phase space

...
come back to
later...

Instead...

- Instead...

II. PLEASE REMEMBER

A. $p \equiv (E, \vec{p})$ is a 4-vector (& masses are usually ~ 0 !)

B. Products of 4-vectors are invariant

Griffiths

C. $\frac{dp_{||}}{E}$ is an invariant, p_T is an invariant

$$\left. \begin{aligned} p_{||} &= m_T \sinh y \\ E &= m_T \cosh y \end{aligned} \right\} \Leftrightarrow \left\{ \begin{aligned} m_T &= \sqrt{m^2 + p_T^2} \\ y &= \frac{1}{2} \ln \frac{E + p_{||}}{E - p_{||}} \end{aligned} \right.$$

E. $\beta = p/E = \tanh y$

F. $E \frac{d^3n}{dp^3} = \frac{d^3n}{dy d^2p_T} = \text{invariant}$

G. Lorentz transforms additive in rapidity

$p^2 = \text{Mass}^2$ Most Important

- This relation is the most important 4-vector dot product
- $p \circ p = p^2 = m^2$
- Any momentum in ME bigger than m_{electron}
- ...and m_{pion} and m_{muon}
- ...but not $m_{J/\Psi}$

Angle Problem: dot products

- This HW2 problem is not like the other one using the vector version of BOOST equation
- Use invariant 4-vectors

3.15 A pion traveling at speed v decays into a muon and a neutrino, $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$. If the neutrino emerges at 90° to the original pion direction, at what angle does the μ come off? [Answer: $\tan\theta = (1 - m_\mu^2/m_\pi^2)/(2\beta\gamma^2)$.]

- (You might try to do problem 2 this way and compare)

$$\pi \rightarrow \mu + \nu$$
$$\overset{\substack{\uparrow \\ \text{4vector}}}{P_\pi^2} = (P_\mu + P_\nu)^2 = P_\mu^2 + P_\nu^2 + 2P_\mu \cdot P_\nu$$
$$m_\pi^2 = m_\mu^2 + \cancel{m_\nu^2} + 2P_\mu \cdot P_\nu$$

↓

Exercise

- Two particles a with mass m collide. One is at rest in the lab. The other has beam energy E and collides with it and produces two heavier particles, each of mass M .
- $a+a \rightarrow b+b$;
- Half the class find expression for four vector p_1 dot p_2 and p_1+p_2 in lab frame “(fixed target)”.
- Half the class find “ ” in the COM frame

In class we mostly focused on the initial state (p_1+p_2)
For Lab frame it was easiest to evaluate p_1 dot p_2 directly
For the COM frame p_1+p_2 (four vector then square it)

The next slides show this done in a different context, at the end of class, I set the two results equal to demonstrate the advantage of using colliders (ie COM frame) or conversely, the much larger beam E needed for the same COM energy in the fixed target

CoM (yes $\equiv$ Collider) frames

- Both result in super trivial stuff – use E_b for beam E
 - Also as long as E 's are big enough doesn't really need to be same mass particles
- 1) Add vectors first :

$$S = \left([E_b, \vec{p}_1] + [E_b, \vec{p}_2] \right)^2 = \left([2E_b, 0] \right)^2 = (2E_b)^2 = 4E_b^2$$

- 2) Do 4-v dot products first

$$S = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = \cancel{m_1^2} + \cancel{m_2^2} + 2p_1 \cdot p_2$$

- tendency forget?: diff 4-v dotted also invariant – can view as item to compare in both frames

$$E_b \quad \xrightarrow{\vec{p}_1 = E_b \hat{z}}$$

$$\xleftarrow{\vec{p}_2 = -E_b \hat{z}} E_b$$

CoM (yes $\equiv$ Collider) frames

- Both result in super trivial stuff – use E_b for beam E
 - Also as long as E 's are big enough doesn't really need to be same mass particles
- 2) Do 4-v dot products first

$$s = p_1^2 + p_2^2 + 2 p_1 \cdot p_2 = \cancel{m_1^2} + \cancel{m_2^2} + 2 p_1 \cdot p_2$$

$2[(m, 0) \cdot (E, \mathbf{p})]$
(in f.t. frame)

$$2 \left([E_b, E_b \hat{z}] \cdot [E_b, -E_b \hat{z}] \right) = 2(E_b^2 - \vec{p}_1 \cdot \vec{p}_2)$$

$$= 2(E_b^2 + E_b^2) = 4 E_b^2$$

$E_b \xrightarrow{\vec{p}_1 = E_b \hat{z}}$
 $\xleftarrow{\vec{p}_2 = -E_b \hat{z}} E_b$

One more time Fixed Target Frame

- let's just re-iterate this one more time : in Fixed target frame:

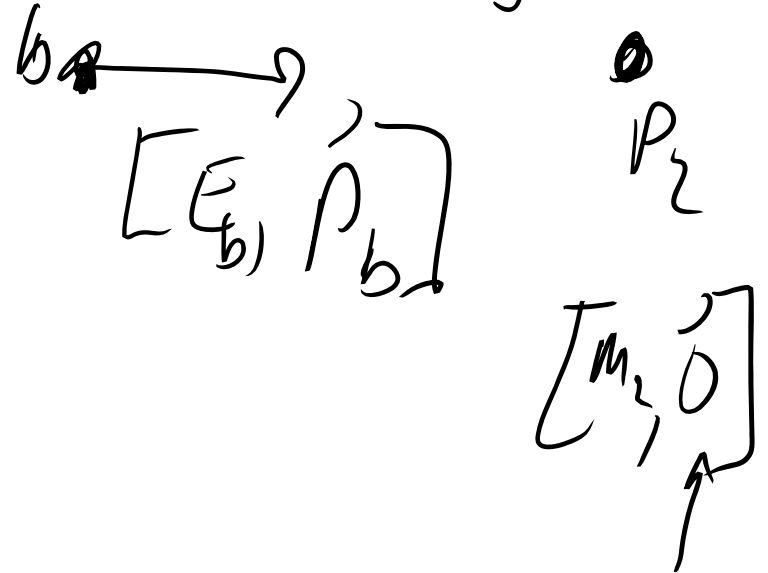
$$s = (p_1 + p_2)^2$$

$$= \cancel{p_1^2} + \cancel{p_2^2} + 2p_1 \cdot p_2$$

$m_1^2 \quad m_2^2$

$$= 2(E_b m_2 - \vec{p}_0 \cdot \vec{0}) = 2E_b m_2$$

$$\sqrt{s} = \sqrt{2E_b m} \quad \sim \quad 2E_{\text{coll, de.b}}$$



IMPLICATION!

- Most important: lab frame vs fixed target
- Large advantage for collider:
- **MUCH HIGHER Ebeam needed for single beam fixed target for the same COM energy**

Fixed Targets vs. Colliding Beams

- In BaBar, $(9 + 3.1) = 12.1$ GeV of beam energy leads to $(2 \times 5.3) = 10.6$ GeV of CM energy that can be used to make new particles (in this case, the $\Upsilon(4S)$).
- How much beam energy would it take to produce this CM energy if the target were fixed?
- Use the total four-momentum as an invariant. The individual four-momenta will be $(m, 0)$ and $(E, \mathbf{p})$, and therefore

$$\begin{aligned}(p_{e^-} + p_{e^-})^2 &= m^2 + m^2 + 2[(m, 0) \cdot (E, \mathbf{p})] \\ &\simeq 2Em\end{aligned}$$

With $2Em = 4E_{CM}^2$ we find that $E = 10^5$ GeV!